

Kyle Hansen

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Curriculum Vitae

Education

- June 2025 **Ph.D. in Mathematics**, *UC Santa Barbara (UCSB)*, Santa Barbara, CA
Advisor: Fedya Manin
Certificate in College and University Teaching (CCUT), *UCSB*, Santa Barbara, CA
- June 2021 **M.A. in Mathematics**, *UCSB*, Santa Barbara, CA
- May 2019 **B.S. in Mathematics**, *Westmont College*, Santa Barbara, CA
B.A. in Computer Science, *Westmont College*, Santa Barbara, CA

Employment

- 2025-Present **Assistant Professor**, *Mathematics*, Westmont College, Santa Barbara, CA
- 2021-2023 **Adjunct Professor**, *Mathematics*, Westmont College, Santa Barbara, CA
- 2023 **Adjunct Professor**, *Computer Science*, UCSB, Santa Barbara, CA
- 2022 **Lead Teaching Assistant**, *Mathematics*, UCSB, Santa Barbara, CA
- 2019-2025 **Teaching Assistant**, *Mathematics*, UCSB, Santa Barbara, CA

Teaching Experience

Instructor of Record

Calculus I, *Westmont College*
F 2026, S 2026

Calculus II, *Westmont College*
S 2026

Discrete Mathematics, *Westmont College*
F 2026, F 2025, F 2023, F 2022, F 2021

Senior Seminar, *Westmont College*
S 2026

Abstract Alegbra, *Westmont College*
F 2025

Foundations of Computer Science, *UCSB*
Sp 2023

Upper Division Teaching Assistant

Linear Algebra, UCSB
Sp 2024, F 2023, W 2022*

Numerical Methods, UCSB
W 2024

Methods of Analysis, *UCSB*
F 2021, Su 2021*, W2025

Lower Division Teaching Assistant

* this term was conducted remotely

Transition to Higher Mathematics, UCSB

Su 2024, Su 2023, W 2021*, Su 2020*, Sp 2020*

Linear Algebra with Applications, UCSB

F 2022, Su 2022, F 2020*

Differential Equations, UCSB

Sp 2021*

Calculus for the Social and Life Sciences I, UCSB

W 2020

Calculus with Applications I, UCSB

F 2019

Course Grader

Modern Euclidean and Noneuclidean Geometry, UCSB

F 2024

Discrete Mathematics, Westmont College

F 2018, F 2017, F 2016

MA-020 Linear Algebra, Westmont College

Sp 2018

Fundamentals of Mathematics II, Westmont College

Sp 2019, Sp 2017

Publications

In Preparation

- [1] Kyle Hansen. Simple tilings of nilpotent Lie groups. *arXiv preprint arXiv:2402.10194*, 2024.

Under Review

- [2] Kyle Hansen. Volumes of nullhomotopies in nilpotent spaces. *arXiv preprint arXiv:2510.00159*, 2025.

Published

- [3] Kyle Hansen. *Volumes of Nullhomotopies in Nilpotent Spaces*. PhD thesis, UCSB, 2025.
- [4] David Hansen and Kyle Hansen. Clues about bluffing in Clue: Is conventional wisdom wise? *IEEE Transactions on Games*, 13(3):310–314, 2019.
- [5] Yury Ionin and Kyle Hansen. A special instance of rock–paper–scissors. *The American Mathematical Monthly*, 125(8):757–758, 2018.

Selected Presentations

- 2026 **Volumes of Nullhomotopies in Nilpotent Spaces**, JMM, Washington DC
- 2025 **Volumes of Nullhomotopies in Nilpotent Spaces**, *AMS SE Sectional Meeting*, Tulane University, New Orleans, LA
- 2025 **Who Cut the High-Dimensional Cheese?**, *Grad Slam*, UCSB, Santa Barbara, CA
- Mathematical Quasicrystals: The Taming of the Hullabaloo**, *Colloquium*, Wheaton College, Wheaton, IL
- 2024 **Tiling Spaces Which are Fiber Bundles over Nilmanifolds**, *41st Workshop in Geometric Topology*, Calvin University, Grand Rapids, MI
- 2022 **The Hull of a Manifold with Bounded Geometry**, *Topology Seminar*, UCSB, Santa Barbara, CA

2019 **When the Fundamental Theorem of Algebra Goes Awry**, *Contributed Paper Session on Revitalizing Complex Analysis*, JMM, Baltimore, MD
 Joint with Mike Brilleslyper (Florida Poly) and Russ Howell (Westmont College)

Research Interests

- Aperiodic Tilings **Percolation on Aperiodic Tilings**, *Westmont College*, Santa Barbara, CA
 Estimate the percolation threshold for aperiodic tilings such as the Penrose Tiling
Collaborators: Erik Potts (Undergraduate Student)
Aperiodic Tilings in Nilpotent Lie Groups, *UCSB*, Santa Barbara, CA
 Determine topological structures of tiling spaces generated by tilings of nilpotent Lie groups
Collaborators: Fedya Manin (Ph.D. Advisor)
- Quantitative Topology **Volumes of Nullhomotopies in Nilpotent Spaces**, *UCSB*, Santa Barbara, CA
 Quantify the volumes of nullhomotopies in nilpotent spaces
Collaborators: Fedya Manin (Ph.D. Advisor)
Nilpotent-Elliptic Manifolds, *UCSB*, Santa Barbara, CA
 Detect nilpotent-ellipticity (in the sense of Gromov) of a manifold from its algebraic data
Collaborators: Fedya Manin (Ph.D. Advisor)
- Complex Analysis **Roots of Complex-Harmonic Polynomials**, *Westmont College*, Santa Barbara, CA
 Count and locate the roots of complex harmonic polynomials
Collaborators: Russell Howell (Undergraduate Advisor)
- Computational Graph Theory **Scale-Invariant Properties of Networks**, *National Security Internship Program at the Pacific Northwest National Lab*, Richland, WA
 Analyze large social networks to detect scale-invariant properties
Collaborators: Jennifer Webster (Project Advisor)

Mentorship, Volunteer, and Community Service

- 2025-2026 **L^AT_EX Basics**, *Westmont College*, Santa Barbara, CA
 Provide students across disciplines with foundational understanding of L^AT_EX for professional typesetting
- 2024 **JMM Chaperone**, *Westmont College*, Santa Barbara, CA
 Drive college students to the Joint Mathematics Meeting, and debrief conference talks
- 2024-2025 **Graduate Christian Encouragement (GraCE) Coordinator**, *Unaffiliated*, Santa Barbara, CA
 Organize a weekly informal gathering of Christian graduate students at UCSB
- 2023-2025 **Directed Reading Program Mentor**, *UCSB*, Santa Barbara, CA
 Mentor undergraduates in elective topics: tiling spaces, cryptography, quantum computing
- 2022-Present **Math Field Day Volunteer**, *Westmont College*, Santa Barbara, CA
 Guide undergraduate volunteers in judging high school “chalk talks” in annual competition
- 2022-2025 **Community Group Discussion Leader**, *Christ Presbyterian Church*, Santa Barbara, CA
 Lead adults in church community group discussions
- 2022-Present **Youth Group Volunteer**, *Christ Presbyterian Church*, Santa Barbara, CA
 Mentor high school students in small groups and individually
- 2022-Present **Quire of Voyces**, *Santa Barbara City College*, Santa Barbara, CA
 Sing in a local classical a capella choir
- 2020-Present **Church Setup and Security**, *Christ Presbyterian Church*, Santa Barbara, CA
 Prepare church building for Sunday worship, and ensure safety of attendees
- 2020-2021 **Elementary School Virtual Tutor**, *Partners in Education*, Santa Barbara, CA
 Assist elementary school teachers in overseeing remote math class activities

Conferences & Workshops

- 2026 **Rhythms of Rest and Responsibility: Managing Commitments in the Early Career**, *MathFest*, Boston, MA
Co-organizer with Rafael Ceja Ayala, Adam Mair, and Charith Silva through Project NExT
Virtue Education Workshop, *Pepperdine*, Malibu, CA
Joint Mathematics Meeting, Washington, DC
- 2025 **AMS SE Sectional Meeting**, *Tulane University*, New Orleans, LA
MathFest, Sacramento, CA
- 2024 **41st Workshop in Geometric Topology**, *Calvin University*, Grand Rapids, MI
Quantitative Topology and Beyond, *Simons Foundation*, New York City, NY
Joint Mathematics Meeting, San Francisco, CA
- 2023 **Geometry and Analysis of Groups and Manifolds**, *Centro De Giorgi*, Pisa, Italy
Sub-Riemannian Geometry and Beyond III, *Centro De Giorgi*, Pisa, Italy
- 2022 **Graduate Student Concentration Week on Metric Geometry**, *Texas A&M Workshop in Analysis and Probability*, College Station, TX
- 2019 **Joint Mathematics Meeting**, Baltimore, MD

Fellowships and Memberships

- 2025 Cohort **Project NExT Fellow**
2025-Present **ACMS**
2025-Present **AMS**
2025-Present **MAA**

Organizations and Discussion Groups

- 2026 **GenAI Discussion Group**, *Westmont College*, Santa Barbara, CA
- 2025-Present **Christian Academic Writers**, *Christ Presbyterian Church Santa Barbara*, Santa Barbara, CA
Organize research and writing sessions with other Christian academic writers across disciplines.

Honors & Awards

- 2024 Department of Mathematics Outstanding Teaching Assistant Award *UCSB*
- 2019 Outstanding Graduate Award in Mathematics *Westmont College*
- 2019 Kenneth Monroe Award *Westmont College*

Technical Skills

Programming

LaTeX, Python, R, SQL, Java, Ruby, C, Racket (Scheme)

Teaching Tools

Canvas, Gradescope, WeBWork, Piazza, iClicker